PCS223 Advanced Computer Networks - Assignment 2

1. Objectives

- To capture and analyze ICMP packets (Ping request and reply)
- To analyze TCP packets and observe the TCP three-way handshake
- To capture and analyze UDP packets
- To capture and analyze DNS query and response packets
- To capture and analyze HTTP packets
- To analyze ARP request and reply packets
- To identify source and destination MAC and IP addresses in captured packets

2. Procedure & Analysis

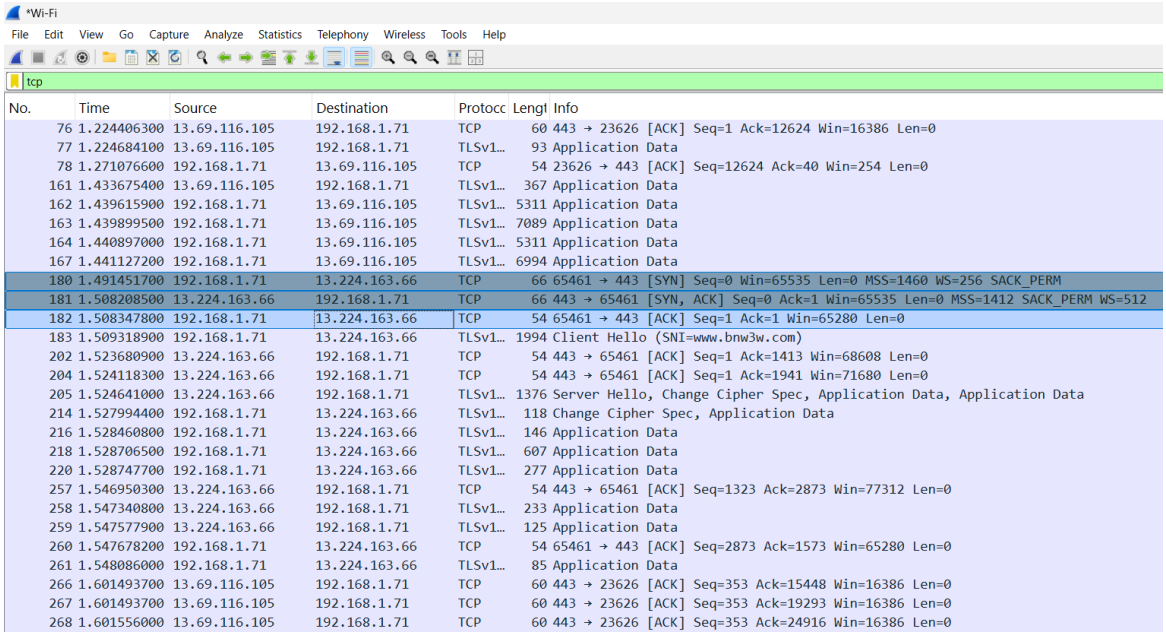
1. ICMP Packet Analysis

- Apply filter: icmp
- Run: ping google.com
- Observe Echo Request and Echo Reply

```
C:\Users\inder>ping google.com

Pinging google.com [2404:6800:4002:819::200e] with 32 bytes of data:
Reply from 2404:6800:4002:819::200e: time=86ms
Reply from 2404:6800:4002:819::200e: time=16ms
Reply from 2404:6800:4002:819::200e: time=73ms
Reply from 2404:6800:4002:819::200e: time=19ms

Ping statistics for 2404:6800:4002:819::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 16ms, Maximum = 86ms, Average = 48ms
```

2. TCP Three-Way Handshake

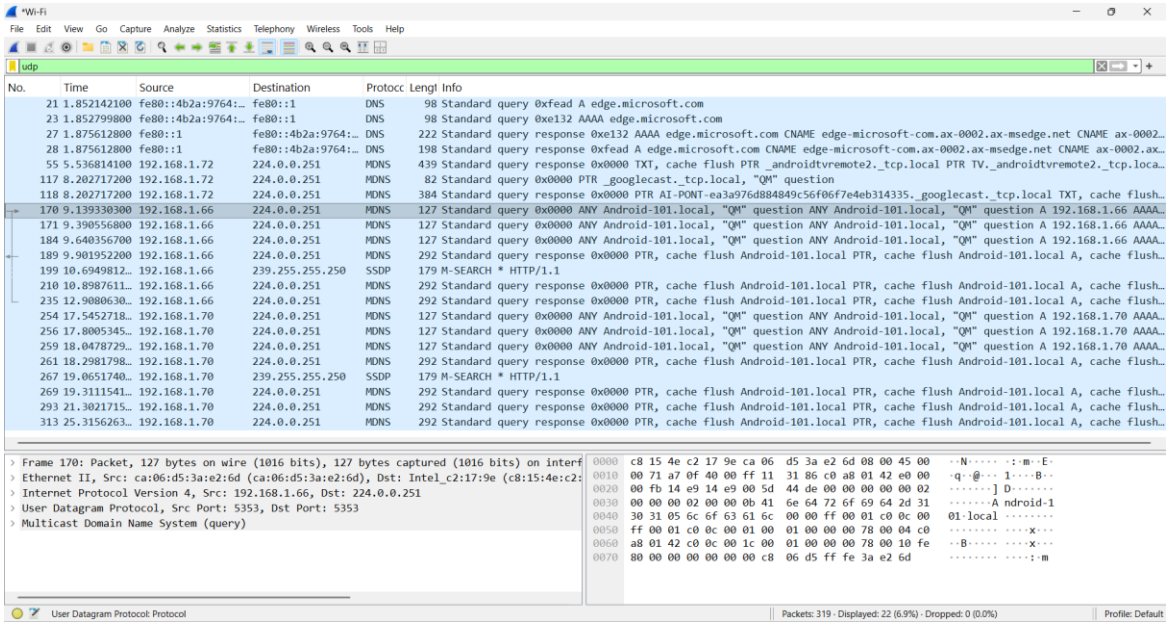
- Apply filter: tcp
- Open a website
- Observe SYN, SYN-ACK, ACK packets



No.	Time	Source	Destination	Protocc	Lengt	Info
76	1.224406300	13.69.116.105	192.168.1.71	TCP	60	443 → 23626 [ACK] Seq=1 Ack=12624 Win=16386 Len=0
77	1.224684100	13.69.116.105	192.168.1.71	TLSv1…	93	Application Data
78	1.271076600	192.168.1.71	13.69.116.105	TCP	54	23626 → 443 [ACK] Seq=12624 Ack=40 Win=254 Len=0
161	1.433675400	13.69.116.105	192.168.1.71	TLSv1…	367	Application Data
162	1.439615900	192.168.1.71	13.69.116.105	TLSv1…	5311	Application Data
163	1.439899500	192.168.1.71	13.69.116.105	TLSv1…	7089	Application Data
164	1.440897000	192.168.1.71	13.69.116.105	TLSv1…	5311	Application Data
167	1.441127200	192.168.1.71	13.69.116.105	TLSv1…	6994	Application Data
180	1.491451700	192.168.1.71	13.224.163.66	TCP	66	65461 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
181	1.508208500	13.224.163.66	192.168.1.71	TCP	66	443 → 65461 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=512
182	1.508347800	192.168.1.71	13.224.163.66	TCP	54	65461 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
183	1.509318900	192.168.1.71	13.224.163.66	TLSv1…	1994	Client Hello (SNI=www.bnw3w.com)
202	1.523680900	13.224.163.66	192.168.1.71	TCP	54	443 → 65461 [ACK] Seq=1 Ack=1413 Win=68608 Len=0
204	1.524118300	13.224.163.66	192.168.1.71	TCP	54	443 → 65461 [ACK] Seq=1 Ack=1941 Win=71680 Len=0
205	1.524641000	13.224.163.66	192.168.1.71	TLSv1…	1376	Server Hello, Change Cipher Spec, Application Data, Application Data
214	1.527994400	192.168.1.71	13.224.163.66	TLSv1…	118	Change Cipher Spec, Application Data
216	1.528460800	192.168.1.71	13.224.163.66	TLSv1…	146	Application Data
218	1.528706500	192.168.1.71	13.224.163.66	TLSv1…	607	Application Data
220	1.528747700	192.168.1.71	13.224.163.66	TLSv1…	277	Application Data
257	1.546950300	13.224.163.66	192.168.1.71	TCP	54	443 → 65461 [ACK] Seq=1323 Ack=2873 Win=77312 Len=0
258	1.547340800	13.224.163.66	192.168.1.71	TLSv1…	233	Application Data
259	1.547577900	13.224.163.66	192.168.1.71	TLSv1…	125	Application Data
260	1.547678200	192.168.1.71	13.224.163.66	TCP	54	65461 → 443 [ACK] Seq=2873 Ack=1573 Win=65280 Len=0
261	1.548086000	192.168.1.71	13.224.163.66	TLSv1…	85	Application Data
266	1.601493700	13.69.116.105	192.168.1.71	TCP	60	443 → 23626 [ACK] Seq=353 Ack=15448 Win=16386 Len=0
267	1.601493700	13.69.116.105	192.168.1.71	TCP	60	443 → 23626 [ACK] Seq=353 Ack=19293 Win=16386 Len=0
268	1.601556000	13.69.116.105	192.168.1.71	TCP	60	443 → 23626 [ACK] Seq=353 Ack=24916 Win=16386 Len=0

3. UDP Packet Analysis

- Apply filter: udp
- Generate traffic (e.g., DNS or streaming apps)
- Observe UDP packets



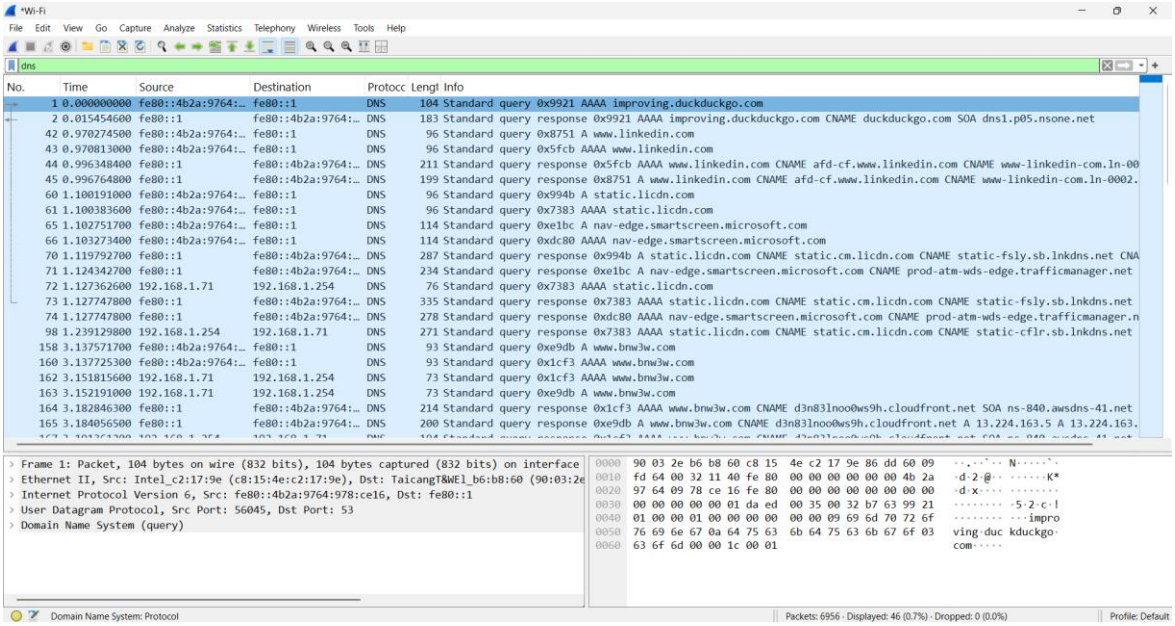
No.	Time	Source	Destination	Protocc	Lengt	Info
21	1.852142100	fe80::4b2a:9764:…	fe80::1	DNS	98	Standard query 0xfead A edge.microsoft.com
23	1.852799800	fe80::4b2a:9764:…	fe80::1	DNS	98	Standard query 0xe132 AAAA edge.microsoft.com
27	1.875612800	fe80::1	fe80::4b2a:9764:…	DNS	222	Standard query response 0xe132 AAAA edge.microsoft.com CNAME edge-microsoft-com.ax-0002.ax-msedge.net CNAME ax-0002…
28	1.875612800	fe80::1	fe80::4b2a:9764:…	DNS	198	Standard query response 0xfead A edge.microsoft.com CNAME edge-microsoft-com.ax-0002.ax-msedge.net CNAME ax-0002.ax…
55	5.536814100	192.168.1.72	224.0.0.251	MDNS	439	Standard query response 0x0000 TXT, cache flush PTR _androidtvremote2._tcp.local PTR TV._androidtvremote2._tcp.loca…
117	8.202717200	192.168.1.72	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
118	8.202717200	192.168.1.72	224.0.0.251	MDNS	384	Standard query response 0x0000 PTR AI-PONT-ea3a976d884849c56f06f7e4eb314335._googlecast._tcp.local TXT, cache flush…
170	9.139330300	192.168.1.66	224.0.0.251	MDNS	127	Standard query 0x0000 ANY Android-101.local, "QM" question ANY Android-101.local, "QM" question A 192.168.1.66 AAAA…
171	9.390556800	192.168.1.66	224.0.0.251	MDNS	127	Standard query 0x0000 ANY Android-101.local, "QM" question ANY Android-101.local, "QM" question A 192.168.1.66 AAAA…
184	9.640356700	192.168.1.66	224.0.0.251	MDNS	127	Standard query 0x0000 ANY Android-101.local, "QM" question ANY Android-101.local, "QM" question A 192.168.1.66 AAAA…
189	9.901952200	192.168.1.66	224.0.0.251	MDNS	292	Standard query response 0x0000 PTR, cache flush Android-101.local PTR, cache flush Android-101.local A, cache flush…
199	10.6949812…	192.168.1.66	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
210	10.8987611…	192.168.1.66	224.0.0.251	MDNS	292	Standard query response 0x0000 PTR, cache flush Android-101.local PTR, cache flush Android-101.local A, cache flush…
235	12.9080630…	192.168.1.66	224.0.0.251	MDNS	292	Standard query response 0x0000 PTR, cache flush Android-101.local PTR, cache flush Android-101.local A, cache flush…
254	17.5452718…	192.168.1.70	224.0.0.251	MDNS	127	Standard query 0x0000 ANY Android-101.local, "QM" question ANY Android-101.local, "QM" question A 192.168.1.70 AAAA…
256	17.8005345…	192.168.1.70	224.0.0.251	MDNS	127	Standard query 0x0000 ANY Android-101.local, "QM" question ANY Android-101.local, "QM" question A 192.168.1.70 AAAA…
259	18.0478729…	192.168.1.70	224.0.0.251	MDNS	127	Standard query 0x0000 ANY Android-101.local, "QM" question ANY Android-101.local, "QM" question A 192.168.1.70 AAAA…
261	18.2981798…	192.168.1.70	224.0.0.251	MDNS	292	Standard query response 0x0000 PTR, cache flush Android-101.local PTR, cache flush Android-101.local A, cache flush…
267	19.0651740…	192.168.1.70	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
269	19.3111541…	192.168.1.70	224.0.0.251	MDNS	292	Standard query response 0x0000 PTR, cache flush Android-101.local PTR, cache flush Android-101.local A, cache flush…
293	21.3021715…	192.168.1.70	224.0.0.251	MDNS	292	Standard query response 0x0000 PTR, cache flush Android-101.local PTR, cache flush Android-101.local A, cache flush…
313	25.3156263…	192.168.1.70	224.0.0.251	MDNS	292	Standard query response 0x0000 PTR, cache flush Android-101.local PTR, cache flush Android-101.local A, cache flush…

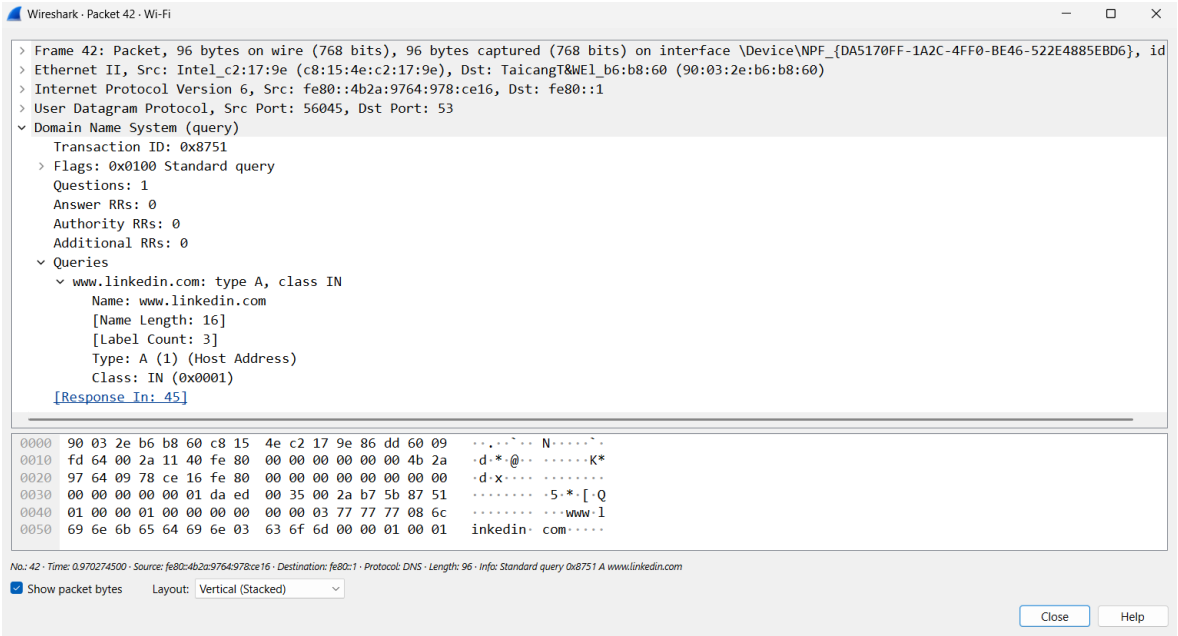
```
> Frame 170: Packet, 127 bytes on wire (1016 bits), 127 bytes captured (1016 bits) on interf
> Ethernet II, Src: ca:06:d5:3a:e2:6d (ca:06:d5:3a:e2:6d), Dst: Intel_c2:17:9e (c8:15:4e:c2:
> Internet Protocol Version 4, Src: 192.168.1.66, Dst: 224.0.0.251
> User Datagram Protocol, Src Port: 5353, Dst Port: 5353
> Multicast Domain Name System (query)

0000  c8 15 4e c2 17 9e ca 06  d5 3a e2 6d 08 00 45 00   ··N····· ·:·m··E·
0010  00 71 a7 0f 40 00 ff 11  31 86 c0 a8 01 42 e0 00   ·q··@··· 1····B··
0020  00 fb 14 e9 14 e9 00 5d  44 de 00 00 00 00 00 02   ·******· D·******
0030  00 00 00 02 00 00 0b 41  6e 64 72 6f 69 64 2d 31   ·******A ndroid-1
0040  30 31 05 6c 6f 63 61 6c  00 00 ff 00 01 c0 0c 00   01·local ********
0050  ff 00 01 c0 0c 00 01 00  01 00 00 00 78 00 04 c0   ******** ····x···
0060  a8 01 42 c0 0c 00 1c 00  01 00 00 00 78 00 10 fe   ··B····· ····x···
0070  80 00 00 00 00 00 00 c8  06 d5 ff fe 3a e2 6d      ******** ····:·m
```

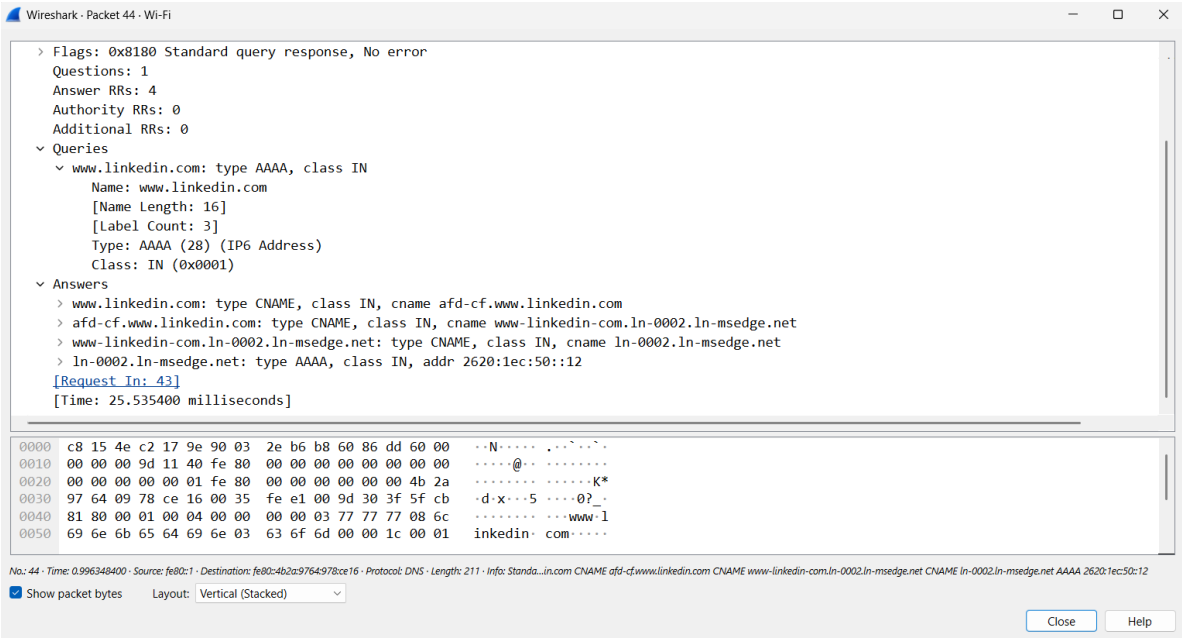
User Datagram Protocol: Protocol | Packets: 319 · Displayed: 22 (6.9%) · Dropped: 0 (0.0%) | Profile: Default

4. DNS Query & Response

- Apply filter: dns
- Visit any website
- Observe query and response packets

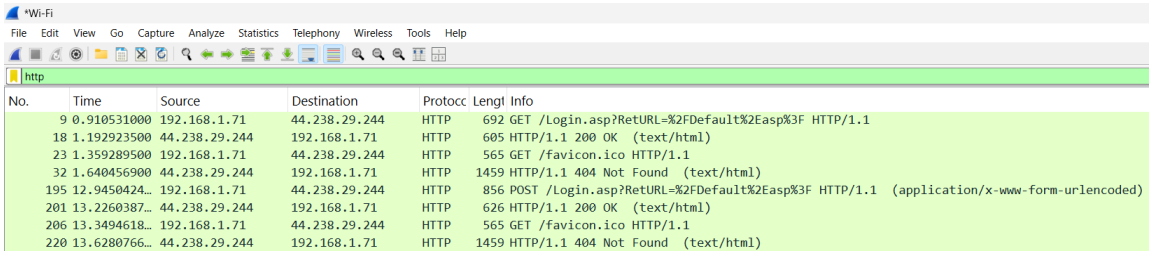

*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocc Leng Info
1 0.000000000 fe80::4b2a:9764:… fe80::1 DNS 104 Standard query 0x9921 AAAA improving.duckduckgo.com
2 0.015454600 fe80::1 fe80::4b2a:9764:… DNS 183 Standard query response 0x9921 AAAA improving.duckduckgo.com CNAME duckduckgo.com SOA dns1.p05.nsone.net
42 0.970274500 fe80::4b2a:9764:… fe80::1 DNS 96 Standard query 0x8751 A www.linkedin.com
43 0.970813000 fe80::4b2a:9764:… fe80::1 DNS 96 Standard query 0x5fcb AAAA www.linkedin.com
44 0.996348400 fe80::1 fe80::4b2a:9764:… DNS 211 Standard query response 0x5fcb AAAA www.linkedin.com CNAME afd-cf.www.linkedin.com CNAME www-linkedin-com.ln-00
45 0.996764800 fe80::1 fe80::4b2a:9764:… DNS 199 Standard query response 0x8751 A www.linkedin.com CNAME afd-cf.www.linkedin.com CNAME www-linkedin-com.ln-0002.
60 1.100191000 fe80::4b2a:9764:… fe80::1 DNS 96 Standard query 0x994b A static.licdn.com
61 1.100383600 fe80::4b2a:9764:… fe80::1 DNS 96 Standard query 0x7383 AAAA static.licdn.com
65 1.102751700 fe80::4b2a:9764:… fe80::1 DNS 114 Standard query 0xe1bc A nav-edge.smartscreen.microsoft.com
66 1.103273400 fe80::4b2a:9764:… fe80::1 DNS 114 Standard query 0xdc80 AAAA nav-edge.smartscreen.microsoft.com
70 1.119792700 fe80::1 fe80::4b2a:9764:… DNS 287 Standard query response 0x994b A static.licdn.com CNAME static.cm.licdn.com CNAME static-fsly.sb.lnkdns.net CNA
71 1.124342700 fe80::1 fe80::4b2a:9764:… DNS 234 Standard query response 0xe1bc A nav-edge.smartscreen.microsoft.com CNAME prod-atm-wds-edge.trafficmanager.net
72 1.127362600 192.168.1.71 192.168.1.254 DNS 76 Standard query 0x7383 AAAA static.licdn.com
73 1.127747800 fe80::1 fe80::4b2a:9764:… DNS 335 Standard query response 0x7383 AAAA static.licdn.com CNAME static.cm.licdn.com CNAME static-fsly.sb.lnkdns.net
74 1.127747800 fe80::1 fe80::4b2a:9764:… DNS 278 Standard query response 0xdc80 AAAA nav-edge.smartscreen.microsoft.com CNAME prod-atm-wds-edge.trafficmanager.n
98 1.239129800 192.168.1.254 192.168.1.71 DNS 271 Standard query response 0x7383 AAAA static.licdn.com CNAME static.cm.licdn.com CNAME static-cflr.sb.lnkdns.net
158 3.137571700 fe80::4b2a:9764:… fe80::1 DNS 93 Standard query 0xe9db A www.bnw3w.com
160 3.137725300 fe80::4b2a:9764:… fe80::1 DNS 93 Standard query 0x1cf3 AAAA www.bnw3w.com
162 3.151815600 192.168.1.71 192.168.1.254 DNS 73 Standard query 0x1cf3 AAAA www.bnw3w.com
163 3.152191000 192.168.1.71 192.168.1.254 DNS 73 Standard query 0xe9db A www.bnw3w.com
164 3.182846300 fe80::1 fe80::4b2a:9764:… DNS 214 Standard query response 0x1cf3 AAAA www.bnw3w.com CNAME d3n83lnoo0ws9h.cloudfront.net SOA ns-840.awsdns-41.net
165 3.184056500 fe80::1 fe80::4b2a:9764:… DNS 200 Standard query response 0xe9db A www.bnw3w.com CNAME d3n83lnoo0ws9h.cloudfront.net A 13.224.163.5 A 13.224.163.
Frame 1: Packet, 104 bytes on wire (832 bits), 104 bytes captured (832 bits) on interface
Ethernet II, Src: Intel_c2:17:9e (c8:15:4e:c2:17:9e), Dst: TaicangT&WEl_b6:b8:60 (90:03:2e
Internet Protocol Version 6, Src: fe80::4b2a:9764:978:ce16, Dst: fe80::1
User Datagram Protocol, Src Port: 56045, Dst Port: 53
Domain Name System (query)
0000 90 03 2e b6 b8 60 c8 15 4e c2 17 9e 86 dd 60 09
0010 fd 64 00 32 11 40 fe 80 00 00 00 00 00 00 4b 2a
0020 97 64 09 78 ce 16 fe 80 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 01 da ed 00 35 00 32 b7 63 99 21
0040 01 00 00 01 00 00 00 00 00 00 09 69 6d 70 72 6f
0050 76 69 6e 67 0a 64 75 63 6b 64 75 63 6b 67 6f 03
0060 63 6f 6d 00 00 1c 00 01
Domain Name System: Protocol
Packets: 6956 · Displayed: 46 (0.7%) · Dropped: 0 (0.0%)
Profile: Default

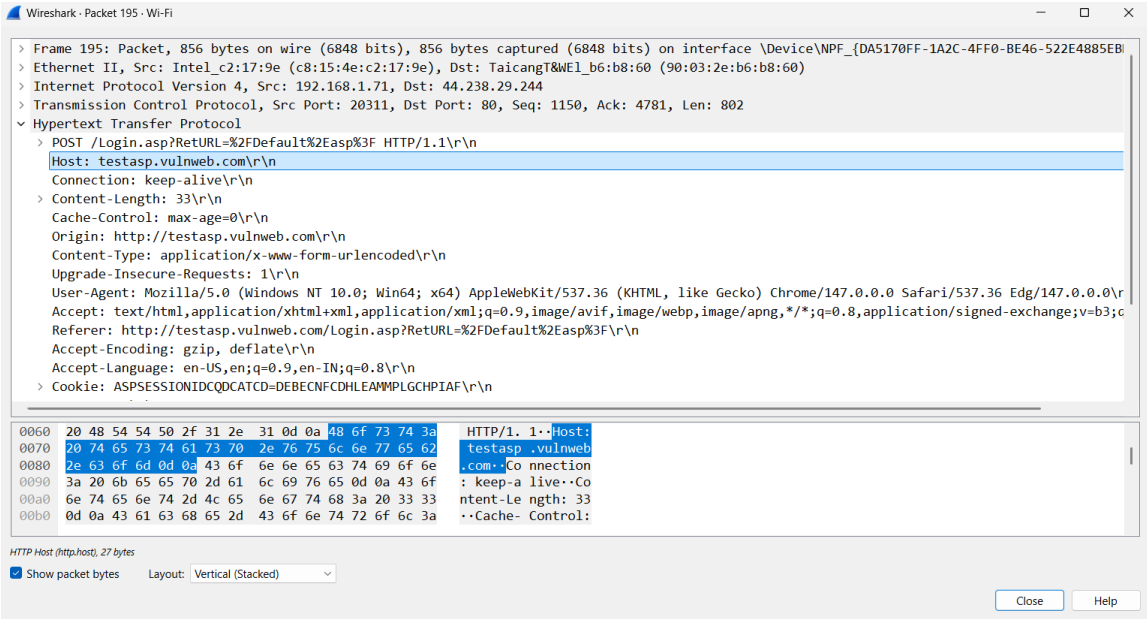

Wireshark · Packet 42 · Wi-Fi
Frame 42: Packet, 96 bytes on wire (768 bits), 96 bytes captured (768 bits) on interface \Device\NPF_{DA5170FF-1A2C-4FF0-BE46-522E4885EBD6}, id
Ethernet II, Src: Intel_c2:17:9e (c8:15:4e:c2:17:9e), Dst: TaicangT&WEl_b6:b8:60 (90:03:2e:b6:b8:60)
Internet Protocol Version 6, Src: fe80::4b2a:9764:978:ce16, Dst: fe80::1
User Datagram Protocol, Src Port: 56045, Dst Port: 53
Domain Name System (query)
Transaction ID: 0x8751
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
www.linkedin.com: type A, class IN
Name: www.linkedin.com
[Name Length: 16]
[Label Count: 3]
Type: A (1) (Host Address)
Class: IN (0x0001)
[Response In: 45]
0000 90 03 2e b6 b8 60 c8 15 4e c2 17 9e 86 dd 60 09
0010 fd 64 00 2a 11 40 fe 80 00 00 00 00 00 00 4b 2a
0020 97 64 09 78 ce 16 fe 80 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 01 da ed 00 35 00 2a b7 5b 87 51
0040 01 00 00 01 00 00 00 00 00 00 03 77 77 77 08 6c
0050 69 6e 6b 65 64 69 6e 03 63 6f 6d 00 00 01 00 01
No.: 42 · Time: 0.970274500 · Source: fe80::4b2a:9764:978:ce16 · Destination: fe80::1 · Protocol: DNS · Length: 96 · Info: Standard query 0x8751 A www.linkedin.com
Show packet bytes
Layout: Vertical (Stacked)
Close
Help

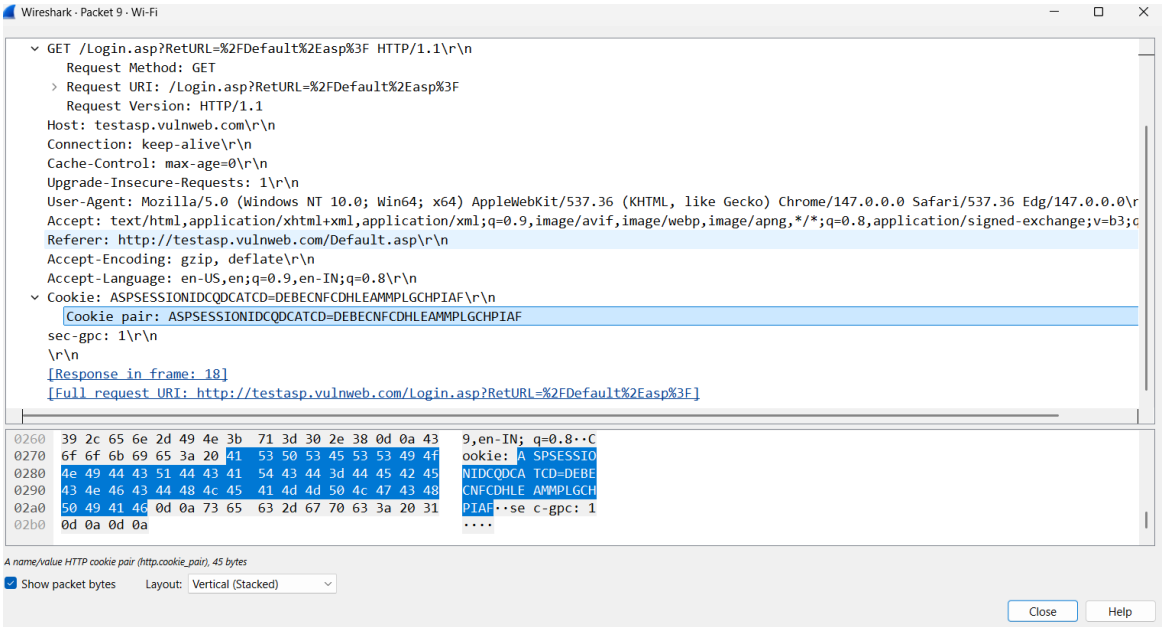


5. HTTP Packet Analysis

- Apply filter: http
- Visit a non-HTTPS website
- Observe request and response







6. ARP Packet Analysis

- Apply filter: arp
- Observe ARP request and reply packets

*Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

arp

No.	Time	Source	Destination	Protocc	Lengt	Info
20	1.153489200	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
22	1.695824900	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
23	1.699547200	TaicangT&WEl_b6:…	Intel_c2:17:9e	ARP	64	Who has 192.168.1.71? Tell 192.168.1.254
24	1.699629200	Intel_c2:17:9e	TaicangT&WEl_b6:…	ARP	42	192.168.1.71 is at c8:15:4e:c2:17:9e
28	2.697030200	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
166	6.986461700	TaicangT&WEl_b6:…	Intel_c2:17:9e	ARP	64	Who has 192.168.1.71? Tell 192.168.1.254
167	6.986549800	Intel_c2:17:9e	TaicangT&WEl_b6:…	ARP	42	192.168.1.71 is at c8:15:4e:c2:17:9e
222	9.153540700	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
253	9.704354200	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
494	10.6961436…	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
1009	12.1160630…	TaicangT&WEl_b6:…	Intel_c2:17:9e	ARP	64	Who has 192.168.1.71? Tell 192.168.1.254
1010	12.1161738…	Intel_c2:17:9e	TaicangT&WEl_b6:…	ARP	42	192.168.1.71 is at c8:15:4e:c2:17:9e
1067	14.0152960…	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
1112	14.6954323…	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
1123	15.6912333…	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
1133	16.5926821…	TaicangT&WEl_b6:…	Intel_c2:17:9e	ARP	64	Who has 192.168.1.71? Tell 192.168.1.254
1134	16.5927492…	Intel_c2:17:9e	TaicangT&WEl_b6:…	ARP	42	192.168.1.71 is at c8:15:4e:c2:17:9e
1136	17.0300374…	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
1152	17.6953264…	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
1173	18.6961368…	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
1214	21.0323033…	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
1450	21.7055573…	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71

2680	129.774136…	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71
2693	130.793941…	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.70? Tell 192.168.1.71
2694	130.797934…	66:e5:29:a0:48:09	Intel_c2:17:9e	ARP	46	192.168.1.70 is at 66:e5:29:a0:48:09
2713	132.139172…	Intel_c2:17:9e	Broadcast	ARP	42	Who has 192.168.1.73? Tell 192.168.1.71

Wireshark · Packet 2693 · Wi-Fi

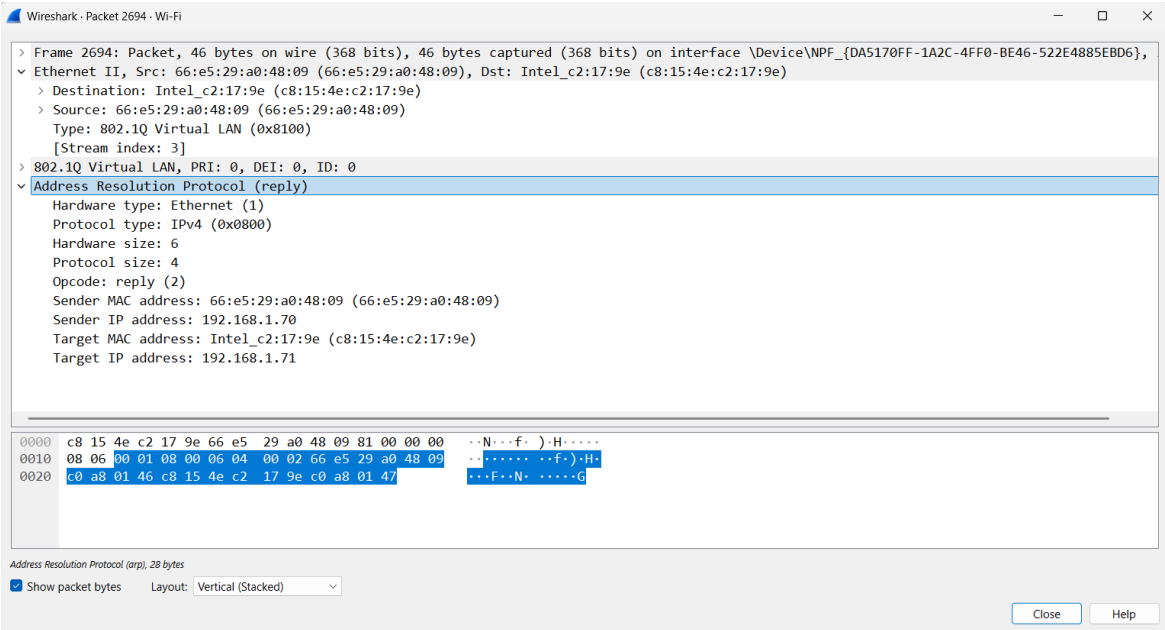
```
> Frame 2693: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{DA5170FF-1A2C-4FF0-BE46-522E4885EBD6},
v Ethernet II, Src: Intel_c2:17:9e (c8:15:4e:c2:17:9e), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  > Destination: Broadcast (ff:ff:ff:ff:ff:ff)
  > Source: Intel_c2:17:9e (c8:15:4e:c2:17:9e)
    Type: ARP (0x0806)
    [Stream index: 1]
v Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: Intel_c2:17:9e (c8:15:4e:c2:17:9e)
    Sender IP address: 192.168.1.71
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 192.168.1.70
```

```
0000  ff ff ff ff ff ff c8 15  4e c2 17 9e 08 06 00 01   ........ N.......
0010  08 00 06 04 00 01 c8 15  4e c2 17 9e c0 a8 01 47   ........ N......G
0020  00 00 00 00 00 00 c0 a8  01 46                     ........ .F
```

No.: 2693 · Time: 130.793941000 · Source: Intel_c2:17:9e · Destination: Broadcast · Protocol: ARP · Length: 42 · Info: Who has 192.168.1.70? Tell 192.168.1.71

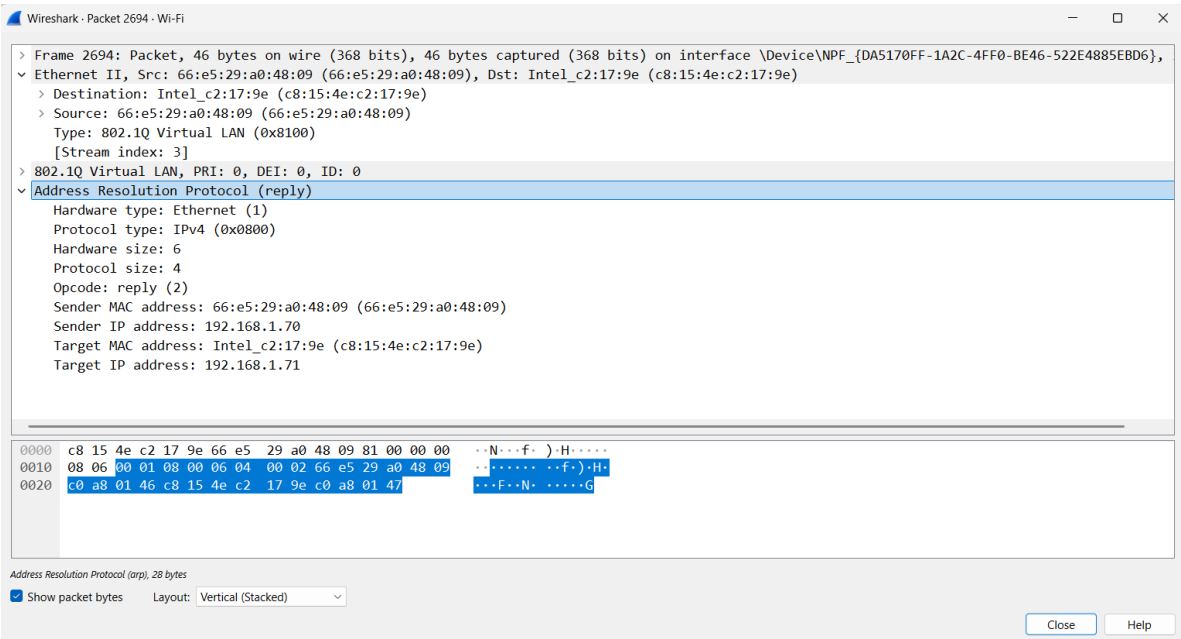
Show packet bytes Layout: Vertical (Stacked)

Close Help



7. MAC & IP Address Identification

- Select any packet
- Note source & destination MAC and IP addresses



Source IP: 192.168.1.70

Source Mac: 66:e5:29:a0:48:09

Destination IP: 192.168.1.71

Destination MAC: c8:15:4e:c2:17:9e

3. Result

All required network packets (ICMP, TCP, UDP, DNS, HTTP, ARP) were successfully captured and analyzed using Wireshark. Source and destination MAC and IP addresses were identified correctly.